

MATA37H3 — Week 03

Integration Techniques I: Substitution & Integration by Parts

Recap and orientation

FTC (Week 2) reduced every definite integral to “find an antiderivative, then subtract endpoint values.” This unit is the first of two building a toolkit for actually *finding* antiderivatives when the integrand isn’t already a basic form. Here we cover the two most fundamental techniques: **substitution** (undoing the chain rule) and **integration by parts** (undoing the product rule). Next week’s session adds trigonometric-integral identities and partial fractions to the toolkit.

Why this matters

Nearly every integral on a MATA37/MAT137-style exam is designed to test whether you can correctly *identify* the technique, not just execute it. Substitution and integration by parts are the two techniques you will reach for constantly — inside trig integrals, inside partial-fraction results, inside improper-integral evaluations, and throughout the series unit when checking that a power series’ antiderivative was computed correctly.

KEY FACT Integration by Substitution (Undoing the Chain Rule) (MAT137 Unit 9)

If F is an antiderivative of f , then $\frac{d}{dx}[F(g(x))] = f(g(x))g'(x)$ by the chain rule, so

$$\int f(g(x)) g'(x) dx = F(g(x)) + C.$$

In practice: pick $u = g(x)$ (the “inside function”), compute $du = g'(x) dx$, and rewrite the whole integral in terms of u only — if any leftover x ’s remain after substituting, the wrong u was chosen (or the integral needs a different technique entirely).

KEY FACT Substitution in a Definite Integral (MAT137 Unit 9)

When evaluating $\int_a^b f(g(x))g'(x) dx$ by substitution $u = g(x)$, the **limits of integration must also change**, to $u = g(a)$ and $u = g(b)$:

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

A common but invalid shortcut is to substitute $u = g(x)$ throughout the integrand while leaving the limits as the original x -values — this produces a numeric answer that is often coincidentally close but is not justified by the change-of-variables theorem, and the *evaluation* step (plugging in numbers) must use whichever variable’s limits you are currently using.

KEY FACT Integration by Parts (OpenStax §3.1, Theorem 3.1)

If $u = f(x)$ and $v = g(x)$ have continuous derivatives, then

$$\int u dv = uv - \int v du.$$

This comes directly from integrating the product rule $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$. Use it when the integrand is a **product** of two functions where one part becomes simpler when differentiated (choose that part as u) and the other part is easy to integrate (that part is dv). A common mnemonic for choosing u is **LIATE** (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential — earlier in the list is generally the better choice of u).

KEY FACT Repeated / Tabular Integration by Parts (MAT137 Unit 9; Dummit)

When the “algebraic” factor is a polynomial of degree $n \geq 2$ (e.g. $\int x^3 e^x dx$), integration by parts must be applied **repeatedly**, each time differentiating the polynomial part further, until it reaches degree 0. The **tabular method** organizes this bookkeeping: make a column of successive derivatives of the polynomial factor (down to 0) alongside a column of successive antiderivatives of the other factor, then combine diagonally with alternating signs. When the pattern instead **cycles back to the original integral** (as with $\int e^{ax} \sin(bx) dx$, where two rounds of parts return a multiple of the same integral), solve for the original integral algebraically rather than trying to apply parts a third time.

Worked Examples

EXAMPLE 1

Basic substitution

Evaluate $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$.

Let $u = \sqrt{x} = x^{1/2}$, so $du = \frac{1}{2\sqrt{x}} dx$, i.e. $\frac{dx}{\sqrt{x}} = 2 du$. Then

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = \int \sin u \cdot 2 du = -2 \cos u + C = -2 \cos \sqrt{x} + C. \quad \blacksquare$$

EXAMPLE 2

Substitution in a definite integral — done correctly

Evaluate $I = \int_0^2 \sqrt{x^3 + 1} x^2 dx$.

Let $u = x^3 + 1$, so $du = 3x^2 dx$, i.e. $x^2 dx = \frac{1}{3} du$. **Changing the limits:** when $x = 0$, $u = 1$; when $x = 2$, $u = 9$. So

$$I = \int_1^9 \sqrt{u} \cdot \frac{1}{3} du = \frac{1}{3} \cdot \frac{2}{3} u^{3/2} \Big|_1^9 = \frac{2}{9} (9^{3/2} - 1^{3/2}) = \frac{2}{9} (27 - 1) = \frac{52}{9}.$$

Note that the limits were changed to u -values (1 and 9), not left as 0 and 2 — mixing the new integrand with the old limits would be evaluating the wrong quantity. $\blacksquare$

EXAMPLE 3

Integration by parts, single application

Evaluate $\int x e^{-2x} dx$.

Choose $u = x$ (gets simpler when differentiated), $dv = e^{-2x} dx$. Then $du = dx$, $v = -\frac{1}{2}e^{-2x}$.
 $\int x e^{-2x} dx = uv - \int v du = -\frac{x}{2}e^{-2x} - \int (-\frac{1}{2}e^{-2x}) dx = -\frac{x}{2}e^{-2x} + \frac{1}{2} \int e^{-2x} dx = -\frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x} + C$. ■

EXAMPLE 4

Repeated integration by parts

Evaluate $\int x^2 \sin x dx$.

First round: $u = x^2$, $dv = \sin x dx \Rightarrow du = 2x dx$, $v = -\cos x$:

$$\int x^2 \sin x dx = -x^2 \cos x + \int 2x \cos x dx.$$

Second round on $\int 2x \cos x dx$: $u = 2x$, $dv = \cos x dx \Rightarrow du = 2 dx$, $v = \sin x$:

$$\int 2x \cos x dx = 2x \sin x - \int 2 \sin x dx = 2x \sin x + 2 \cos x + C.$$

Combining, $\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$. ■

EXAMPLE 5

Tabular integration by parts — three rounds in one table

Evaluate $\int x^3 e^x dx$ using the tabular method.

Differentiate x^3 down to 0 in one column, and antidifferentiate e^x (which reproduces itself) in a parallel column, alternating signs $+, -, +, -$ down the diagonal products:

Sign	D -column (derivatives of x^3)	I -column (antiderivatives of e^x)
+	x^3	e^x
−	$3x^2$	e^x
+	$6x$	e^x
−	6	e^x
	0	e^x

Multiply each D -entry by the *next* I -entry down, with the listed sign, and stop once the D -column reaches 0:

$$\int x^3 e^x dx = x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C.$$

This is exactly three applications of integration by parts done in one pass — the tabular method is a bookkeeping shortcut, not a new theorem, and it only works cleanly when one column terminates at 0 (a polynomial) while the other column is easy to keep antidifferentiating (as e^x , $\sin x$, $\cos x$ all are). ■

EXAMPLE 6

The “solve for the integral” trick — a column that never terminates

Evaluate $I = \int e^x \sin x dx$.

Round 1: $u = e^x$, $dv = \sin x dx \Rightarrow du = e^x dx$, $v = -\cos x$:

$$I = -e^x \cos x + \int e^x \cos x dx.$$

Round 2, on $\int e^x \cos x dx$: $u = e^x$, $dv = \cos x dx \Rightarrow du = e^x dx$, $v = \sin x$:

$$\int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx = e^x \sin x - I.$$

Substituting back: $I = -e^x \cos x + e^x \sin x - I$, so $2I = e^x(\sin x - \cos x)$, giving

$$I = \int e^x \sin x \, dx = \frac{e^x(\sin x - \cos x)}{2} + C.$$

Here neither column of a tabular setup would ever reach 0 (differentiating/antidifferentiating e^x and $\sin x/\cos x$ cycles forever) — this is the signal to stop applying parts and instead solve the resulting algebraic equation for I . ■

EXAMPLE 7

Logarithm as the u — nothing else to differentiate

Evaluate $\int \ln x \, dx$.

Treat this as $\int 1 \cdot \ln x \, dx$: choose $u = \ln x$ (it must be u , since there is no antiderivative shortcut for $dv = \ln x \, dx$), $dv = dx$. Then $du = \frac{1}{x} dx$, $v = x$:

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C. \quad \blacksquare$$

EXAMPLE 8

LIATE with two genuinely different factors

Evaluate $\int x \arctan x \, dx$.

By LIATE, Inverse-trig outranks Algebraic, so $u = \arctan x$, $dv = x \, dx \Rightarrow du = \frac{1}{1+x^2} dx$,

$$v = \frac{x^2}{2}:$$

$$\int x \arctan x \, dx = \frac{x^2}{2} \arctan x - \int \frac{x^2}{2(1+x^2)} \, dx.$$

Simplify the leftover integral by dividing out **before** reaching for partial fractions: $\frac{x^2}{1+x^2} =$

$$1 - \frac{1}{1+x^2}, \text{ so}$$

$$\int x \arctan x \, dx = \frac{x^2}{2} \arctan x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2}\right) dx = \frac{x^2}{2} \arctan x - \frac{x}{2} + \frac{1}{2} \arctan x + C. \quad \blacksquare$$

EXAMPLE 9

Level: Foundational — a trig antiderivative building on Practice Problem 2's cousin

Evaluate $\int \tan x \, dx$.

Rewrite $\tan x = \frac{\sin x}{\cos x}$ and let $u = \cos x$, so $du = -\sin x \, dx$, i.e. $\sin x \, dx = -du$. Then

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx = \int \frac{-du}{u} = -\ln |u| + C = -\ln |\cos x| + C = \ln |\sec x| + C.$$

This is the “twin” of Practice Problem 2 ($\cot x$) — same substitution idea, opposite trig ratio — and the two answers are easy to mix up if the sign or the inside function gets copied on autopilot. ■

EXAMPLE 10

Level: Intermediate — substitution with algebraic rewriting first

Evaluate $\int x^3 \sqrt{x^2 + 1} dx$.

The obvious substitution $u = x^2 + 1$ gives $du = 2x dx$, which only accounts for **one** factor of x — but the integrand has $x^3 = x^2 \cdot x$. Solve for the leftover x^2 in terms of u : since $u = x^2 + 1$, $x^2 = u - 1$. Then

$$x^3 \sqrt{x^2 + 1} dx = x^2 \sqrt{x^2 + 1} \cdot x dx = (u - 1) \sqrt{u} \cdot \frac{du}{2}.$$

$\int x^3 \sqrt{x^2 + 1} dx = \frac{1}{2} \int (u^{3/2} - u^{1/2}) du = \frac{1}{2} \left(\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right) + C = \frac{1}{5} (x^2 + 1)^{5/2} - \frac{1}{3} (x^2 + 1)^{3/2} + C$.
The key move is substituting for the *leftover* power of x in terms of u , rather than declaring the integral “un-doable by substitution” the moment more than one factor of x remains after forming du . ■

EXAMPLE 11

Level: Advanced — integration by parts on a definite integral

Evaluate $\int_0^{\pi/2} x \cos x dx$.

Choose $u = x$, $dv = \cos x dx \Rightarrow du = dx$, $v = \sin x$:

$$\begin{aligned} \int_0^{\pi/2} x \cos x dx &= \left[x \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} \sin x dx = \left(\frac{\pi}{2} \cdot 1 - 0 \right) - \left[-\cos x \right]_0^{\pi/2} \\ &= \frac{\pi}{2} - (-\cos(\pi/2) + \cos 0) = \frac{\pi}{2} - (0 + 1) = \frac{\pi}{2} - 1. \end{aligned}$$

When integration by parts meets a definite integral, the boundary term $[uv]_a^b$ is evaluated at the endpoints **immediately**, and only the remaining integral $\int v du$ carries limits forward — there is no need to find a general antiderivative first and evaluate everything at the very end. ■

EXAMPLE 12

Level: Challenge — substitution feeds into tabular parts

Evaluate $\int x^5 e^{x^2} dx$.

Neither technique works alone: substitution’s obvious candidate $u = x^2$ leaves a leftover x^4 that doesn’t match any single du , and parts with x^5 as the polynomial factor would need five rounds against e^{x^2} — whose antiderivative isn’t even elementary, so parts cannot get started at all. Instead substitute on **part** of the integrand: let $u = x^2$, $du = 2x dx$, so $x^4 = u^2$ and $x dx = \frac{1}{2} du$:

$$\int x^5 e^{x^2} dx = \int x^4 e^{x^2} \cdot x dx = \int u^2 e^u \cdot \frac{du}{2} = \frac{1}{2} \int u^2 e^u du.$$

Now $\int u^2 e^u du$ is a textbook tabular-parts integral (compare Example 5): D -column $u^2, 2u, 2, 0$; I -column e^u, e^u, e^u, e^u ; signs $+, -, +$:

$$\begin{aligned} \int u^2 e^u du &= u^2 e^u - 2u e^u + 2e^u + C. \\ \int x^5 e^{x^2} dx &= \frac{1}{2} (x^4 e^{x^2} - 2x^2 e^{x^2} + 2e^{x^2}) + C = \frac{1}{2} e^{x^2} (x^4 - 2x^2 + 2) + C. \end{aligned}$$

The diagnostic skill here is recognizing that a *partial* substitution — one that simplifies the exponent while deliberately leaving a polynomial factor behind — can turn an integral that resists both techniques individually into one that yields cleanly to parts. ■

Practice Problems

Worked-solutions version: every step of each solution is shown in green beneath its problem.

Substitution

1. Evaluate $\int e^x \cos(e^x) dx$.

SOLUTION

Step 1. Let $u = e^x$, so $du = e^x dx$ — this is exactly the leftover factor in the integrand.

Step 2. Rewrite: $\int e^x \cos(e^x) dx = \int \cos u du$. **Step 3.** Antidifferentiate and substitute back: $\int \cos u du = \sin u + C = \sin(e^x) + C$.

Common misconception: Forgetting that $d(e^x) = e^x dx$ exactly matches the leftover factor — trying an unnecessary second substitution instead of recognizing the integral is already complete after $u = e^x$.

2. Evaluate $\int \cot x dx$.

SOLUTION

Step 1. Rewrite $\cot x = \frac{\cos x}{\sin x}$.

Step 2. Let $u = \sin x$, so $du = \cos x dx$ — again exactly the leftover factor.

Step 3. Rewrite and integrate: $\int \frac{\cos x}{\sin x} dx = \int \frac{du}{u} = \ln |u| + C = \ln |\sin x| + C$.

Common misconception: Confusing $\cot x$'s antiderivative with $\tan x$'s ($-\ln |\cos x| + C$) — these are mirror images and easy to swap.

3. Evaluate $\int_0^1 x e^{-x^2} dx$, carefully converting the limits of integration.

SOLUTION

Step 1. Let $u = x^2$, so $du = 2x dx$, i.e. $x dx = \frac{1}{2} du$.

Step 2. Convert the limits: $x = 0 \Rightarrow u = 0$; $x = 1 \Rightarrow u = 1$.

Step 3. Rewrite entirely in u : $\int_0^1 x e^{-x^2} dx = \frac{1}{2} \int_0^1 e^{-u} du$.

Step 4. Antidifferentiate and evaluate: $\frac{1}{2} [-e^{-u}]_0^1 = \frac{1}{2} (-e^{-1} - (-1)) = \frac{1}{2} (1 - e^{-1})$.

Common misconception: Forgetting to convert the limits from x -values to u -values, and instead plugging the original $x = 0, 1$ bounds into the antiderivative written in terms of u .

4. The following write-up contains an error even though its final numeric answer happens to be wrong: “ $\int_0^2 \sqrt{x^3 + 1} x^2 dx$: let $u = x^3 + 1$, $du = 3x^2 dx$; $= \frac{1}{3} \int_0^2 \sqrt{u} du = \frac{2}{9} u^{3/2} \Big|_0^2$.” Identify the specific error and compute the correct value.

SOLUTION

Step 1. Locate the flaw: the write-up substitutes $u = x^3 + 1$ inside the integrand but never converts the limits $x = 0, 2$ into u -values.

Step 2. Compute the correct limits: $x = 0 \Rightarrow u = 1$ (not $u = 0$); $x = 2 \Rightarrow u = 9$.

Step 3. Redo the evaluation with the correct limits (matching Worked Example 2):

$$\int_1^9 \sqrt{u} \cdot \frac{1}{3} du = \frac{2}{9} (9^{3/2} - 1^{3/2}) = \frac{2}{9} (27 - 1) = \frac{52}{9}.$$

Step 4. Compare: the flawed write-up’s $\frac{2}{9}(2^{3/2} - 0) = \frac{2^{5/2}}{9} \approx 0.63$ is not even close to the correct $\frac{52}{9} \approx 5.78$ — despite the problem statement’s framing, the flawed method does *not* land on the right number by luck here, which is itself worth pointing out to a student who assumes a “close enough” answer signals a minor slip.

Common misconception: Assuming that because a substitution technically “worked” once in a similar-looking example, skipping the limit-conversion step is a safe shortcut — it is not, and can silently change the answer.

5. Evaluate $\int \frac{x}{\sqrt{4 - x^2}} dx$.

SOLUTION

Step 1. Let $u = 4 - x^2$, so $du = -2x dx$, i.e. $x dx = -\frac{1}{2} du$.

Step 2. Rewrite: $\int \frac{x}{\sqrt{4 - x^2}} dx = -\frac{1}{2} \int u^{-1/2} du$.

Step 3. Antidifferentiate: $-\frac{1}{2} \cdot 2u^{1/2} + C = -\sqrt{u} + C = -\sqrt{4 - x^2} + C$.

Common misconception: Reaching for trig substitution ($x = 2 \sin \theta$) when a much simpler direct substitution $u = 4 - x^2$ already matches the leftover $x dx$ factor exactly.

Integration by Parts — single and repeated

6. Evaluate $\int x e^{3x} dx$.

SOLUTION

Step 1. Choose $u = x$, $dv = e^{3x} dx$, so $du = dx$, $v = \frac{1}{3} e^{3x}$.

Step 2. Apply parts: $\int x e^{3x} dx = \frac{x}{3} e^{3x} - \int \frac{1}{3} e^{3x} dx$.

Step 3. Finish the remaining integral: $\frac{x}{3} e^{3x} - \frac{1}{9} e^{3x} + C$.

Common misconception: Forgetting the $\frac{1}{3}$ that comes from antidifferentiating e^{3x} (writing $v = e^{3x}$ instead of $\frac{1}{3}e^{3x}$).

7. Evaluate $\int x^2 \ln x \, dx$.

SOLUTION

Step 1. By LIATE, choose $u = \ln x$ (Logarithmic outranks Algebraic), $dv = x^2 dx$, so $du = \frac{1}{x} dx$, $v = \frac{x^3}{3}$.

Step 2. Apply parts: $\int x^2 \ln x \, dx = \frac{x^3}{3} \ln x - \int \frac{x^3}{3} \cdot \frac{1}{x} dx$.

Step 3. Simplify and finish the remaining integral: $\frac{x^3}{3} \ln x - \int \frac{x^2}{3} dx = \frac{x^3}{3} \ln x - \frac{x^3}{9} + C$.

Common misconception: Choosing $u = x^2$ instead of $u = \ln x$ (violating LIATE) — on this particular integral the leftover terms still simplify, but on harder log-times-polynomial integrals the wrong choice causes the degree of the polynomial factor to increase instead of decrease, and the process never terminates.

8. Evaluate $\int x^4 e^{-x} \, dx$ using the tabular method from Example 5.

SOLUTION

Step 1. Set up the tabular method as in Worked Example 5: differentiate x^4 down to 0 across five rows: $x^4, 4x^3, 12x^2, 24x, 24, 0$.

Step 2. Antidifferentiate e^{-x} repeatedly, tracking the extra sign flip each round (since $\int e^{-x} dx = -e^{-x}$): the I -column is $-e^{-x}, e^{-x}, -e^{-x}, e^{-x}, -e^{-x}$.

Step 3. Combine diagonally with the tabular method's own alternating signs $+, -, +, -, +$: the five products are $-x^4 e^{-x}, -4x^3 e^{-x}, -12x^2 e^{-x}, -24x e^{-x}, -24e^{-x}$.

Step 4. Sum: $\int x^4 e^{-x} dx = -x^4 e^{-x} - 4x^3 e^{-x} - 12x^2 e^{-x} - 24x e^{-x} - 24e^{-x} + C$.

Common misconception: Forgetting that antidifferentiating e^{-x} repeatedly alternates an *extra* sign on top of the tabular method's own alternating signs — losing track of this compounding sign is the single most common tabular-method arithmetic error.

9. Evaluate $\int e^{ax} \cos(bx) \, dx$ (constants $a, b \neq 0$), using the “solve for the integral” trick from Worked Example 6.

SOLUTION

Step 1. Round 1: $u = e^{ax}$, $dv = \cos(bx) dx \Rightarrow du = ae^{ax} dx$, $v = \frac{1}{b} \sin(bx)$:

$$I = \frac{e^{ax} \sin(bx)}{b} - \frac{a}{b} \int e^{ax} \sin(bx) \, dx.$$

Step 2. Round 2 on the remaining integral: $u = e^{ax}$, $dv = \sin(bx)dx \Rightarrow du = ae^{ax}dx$, $v = -\frac{1}{b}\cos(bx)$: $\int e^{ax}\sin(bx)dx = -\frac{e^{ax}\cos(bx)}{b} + \frac{a}{b}\int e^{ax}\cos(bx)dx = -\frac{e^{ax}\cos(bx)}{b} + \frac{a}{b}I$.

Step 3. Substitute back into Step 1: $I = \frac{e^{ax}\sin(bx)}{b} - \frac{a}{b}\left(-\frac{e^{ax}\cos(bx)}{b} + \frac{a}{b}I\right) = \frac{e^{ax}\sin(bx)}{b} + \frac{ae^{ax}\cos(bx)}{b^2} - \frac{a^2}{b^2}I$.

Step 4. Solve for I : $I\left(1 + \frac{a^2}{b^2}\right) = \frac{be^{ax}\sin(bx) + ae^{ax}\cos(bx)}{b^2}$, so $I \cdot \frac{a^2 + b^2}{b^2} = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{b^2}$, giving $I = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{a^2 + b^2} + C$.

Common misconception: Applying integration by parts correctly for two rounds but then trying a third round instead of recognizing the original integral has reappeared and can be solved for algebraically.

Mixed / Applications

10. Estimate $\int_0^1 f'(x)dx$ and $\int_0^3 xf'(x)dx$ from a graph where f is a smooth increasing convex function with $f(0) = 2$, $f(1) = 3$, $f(3) \approx 4.7$ (using $\int_0^1 f'(x)dx = f(1) - f(0)$ via FTC Part 2, and integration by parts on the second one).

SOLUTION

Step 1. For the first integral, apply FTC Part 2 directly: $\int_0^1 f'(x)dx = f(1) - f(0) = 3 - 2 = 1$. **Step 2.** For the second integral, apply parts with $u = x$, $dv = f'(x)dx \Rightarrow du = dx$, $v = f(x)$: $\int_0^3 xf'(x)dx = [xf(x)]_0^3 - \int_0^3 f(x)dx$.

Step 3. Evaluate the boundary term: $[xf(x)]_0^3 = 3f(3) - 0 \cdot f(0) \approx 3(4.7) = 14.1$.

Step 4. So $\int_0^3 xf'(x)dx \approx 14.1 - \int_0^3 f(x)dx$, where $\int_0^3 f(x)dx$ must be read off as the area under the graph of f itself (not f') — this is an estimation problem, so the final numeric value depends on that area reading, but the *setup* above is the exact, non-negotiable target.

Common misconception: Trying to estimate $\int_0^3 xf'(x)dx$ directly from the graph of f (not f') without recognizing integration by parts converts it into terms involving f itself, which the graph of f can estimate.

11. Evaluate $\int \sin(\ln x) dx$.

SOLUTION

Step 1. Round 1: $u = \sin(\ln x)$, $dv = dx \Rightarrow du = \cos(\ln x) \cdot \frac{1}{x} dx$, $v = x$:
 $I = x \sin(\ln x) - \int x \cdot \cos(\ln x) \cdot \frac{1}{x} dx = x \sin(\ln x) - \int \cos(\ln x) dx$.

Step 2. Round 2 on $\int \cos(\ln x) dx$: $u = \cos(\ln x)$, $dv = dx \Rightarrow du = -\sin(\ln x) \cdot \frac{1}{x} dx$, $v = x$: $\int \cos(\ln x) dx = x \cos(\ln x) + \int \sin(\ln x) dx = x \cos(\ln x) + I$.

Step 3. Substitute back into Step 1: $I = x \sin(\ln x) - (x \cos(\ln x) + I) = x \sin(\ln x) - x \cos(\ln x) - I$.

Step 4. Solve for I : $2I = x \sin(\ln x) - x \cos(\ln x)$, so $I = \frac{x(\sin(\ln x) - \cos(\ln x))}{2} + C$.

Common misconception: Trying a substitution $u = \ln x$ first — which does simplify the *argument* but leaves an extra $x = e^u$ factor from $dv = dx$ that must still be integrated by parts — instead of going directly to integration by parts on the original variable.

12. A student claims that $\int \frac{1}{x} \ln x dx$ “obviously needs integration by parts since it’s a product of $\frac{1}{x}$ and $\ln x$.” Show that a substitution is actually faster, and evaluate the integral both ways to confirm they agree.

SOLUTION

Step 1. Substitution route: let $u = \ln x$, $du = \frac{1}{x} dx$: $\int \frac{\ln x}{x} dx = \int u du = \frac{u^2}{2} + C = \frac{(\ln x)^2}{2} + C$.

Step 2. Parts route, for comparison: let $u = \ln x$, $dv = \frac{1}{x} dx \Rightarrow du = \frac{1}{x} dx$, $v = \ln x$:
 $\int \frac{\ln x}{x} dx = (\ln x)^2 - \int \frac{\ln x}{x} dx$.

Step 3. Solve the parts equation for the integral: $2 \int \frac{\ln x}{x} dx = (\ln x)^2$, so
 $\int \frac{\ln x}{x} dx = \frac{(\ln x)^2}{2} + C$ — the same answer as Step 1, but only after the extra algebra of solving for the integral.

Step 4. Conclusion: both methods agree, but substitution reaches the answer in one line while parts requires recognizing and resolving a self-referential equation — substitution is the faster route here.

Common misconception: Assuming integration by parts is always necessary for any integrand that “looks like a product” — $\frac{1}{x} \ln x$ is really $f(g(x))g'(x)$ in disguise (with $g(x) = \ln x$), which substitution handles in one line.

More Practice, Graded by Level and Idea

13. (Level: Foundational — recognition) Evaluate $\int \sec^2(3x) dx$.

SOLUTION

Step 1. Let $u = 3x$, so $du = 3 dx$, i.e. $dx = \frac{1}{3}du$.

Step 2. Rewrite and integrate: $\int \sec^2(3x) dx = \frac{1}{3} \int \sec^2 u du = \frac{1}{3} \tan u + C$.

Step 3. Substitute back: $\frac{1}{3} \tan(3x) + C$.

Common misconception: Writing $\tan(3x) + C$ with no compensating $\frac{1}{3}$ — forgetting that scaling the *inside* of a function by 3 requires dividing the antiderivative by that same 3.

14. (Level: Foundational — inverse trig via parts) Evaluate $\int \arcsin x dx$.

SOLUTION

Step 1. Treat this as $\int 1 \cdot \arcsin x dx$: choose $u = \arcsin x$, $dv = dx$, so $du = \frac{1}{\sqrt{1-x^2}}dx$, $v = x$.

Step 2. Apply parts: $\int \arcsin x dx = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx$.

Step 3. Evaluate the remaining integral by substitution: let $w = 1 - x^2$, $dw = -2x dx$: $\int \frac{x}{\sqrt{1-x^2}} dx = -\frac{1}{2} \int w^{-1/2} dw = -\sqrt{w} + C = -\sqrt{1-x^2} + C$.

Step 4. Combine: $\int \arcsin x dx = x \arcsin x - (-\sqrt{1-x^2}) + C = x \arcsin x + \sqrt{1-x^2} + C$.

Common misconception: Trying $dv = \arcsin x dx$ instead of $u = \arcsin x$ — there is no elementary antiderivative to use as v that way, so $\arcsin x$ (and any inverse trig function with no easy antiderivative) must always be chosen as u , per LIATE.

15. (Level: Intermediate) Evaluate $\int_1^e \ln x dx$ — the definite-integral version of Worked Example 7.

SOLUTION

Step 1. From Worked Example 7, the general antiderivative is $\int \ln x dx = x \ln x - x + C$.

Step 2. Evaluate at the upper limit: $e \ln e - e = e(1) - e = 0$.

Step 3. Evaluate at the lower limit: $1 \ln 1 - 1 = 1(0) - 1 = -1$.

Step 4. Subtract: $\int_1^e \ln x dx = 0 - (-1) = 1$.

Common misconception: Forgetting that $\ln 1 = 0$ (not undefined or 1), which causes the lower-limit evaluation to be miscomputed as $0 - 1 = -1$ read incorrectly,

or dropped entirely.

16. (Level: Intermediate — algebra before technique) Evaluate $\int \frac{x^3}{x^2 + 1} dx$.

SOLUTION

Step 1. Recognize this is not yet in substitution or parts form: the numerator's degree (3) is higher than the denominator's (2), so divide first. Polynomial long division gives $\frac{x^3}{x^2 + 1} = x - \frac{x}{x^2 + 1}$.

Step 2. Split the integral: $\int \frac{x^3}{x^2 + 1} dx = \int x dx - \int \frac{x}{x^2 + 1} dx$.

Step 3. The first piece is immediate: $\int x dx = \frac{x^2}{2}$. The second is a direct substitution, $u = x^2 + 1$, $du = 2x dx$: $\int \frac{x}{x^2 + 1} dx = \frac{1}{2} \ln(x^2 + 1)$.

Step 4. Combine: $\int \frac{x^3}{x^2 + 1} dx = \frac{x^2}{2} - \frac{1}{2} \ln(x^2 + 1) + C$.

Common misconception: Trying to force a substitution $u = x^2 + 1$ on the *whole* fraction right away — it only cleanly matches after the algebraic division step separates out the piece substitution can actually handle.

17. (Level: Advanced) Evaluate $\int x \cos(2x) dx$ by parts, then verify your answer by differentiating it.

SOLUTION

Step 1. Choose $u = x$, $dv = \cos(2x) dx$, so $du = dx$, $v = \frac{1}{2} \sin(2x)$.

Step 2. Apply parts: $\int x \cos(2x) dx = \frac{x}{2} \sin(2x) - \int \frac{1}{2} \sin(2x) dx$.

Step 3. Finish the remaining integral: $\frac{x}{2} \sin(2x) - \frac{1}{2} \left(-\frac{1}{2} \cos(2x)\right) + C = \frac{x}{2} \sin(2x) + \frac{1}{4} \cos(2x) + C$.

Step 4. Verify by differentiating: $\frac{d}{dx} \left[\frac{x}{2} \sin(2x) \right] = \frac{1}{2} \sin(2x) + x \cos(2x)$, and $\frac{d}{dx} \left[\frac{1}{4} \cos(2x) \right] = -\frac{1}{2} \sin(2x)$; summing gives $x \cos(2x)$, matching the original integrand exactly.

Common misconception: Forgetting the extra $\frac{1}{2}$ that appears when antidifferentiating $\int \sin(2x) dx$, and writing $-\cos(2x)$ instead of $-\frac{1}{2} \cos(2x)$.

18. (Level: Advanced — diagnose the error) A student evaluating $\int x e^{x^2} dx$ writes: "Let $u = x$, $dv = e^{x^2} dx$; then $v = e^{x^2}$, so $\int x e^{x^2} dx = x e^{x^2} - \int e^{x^2} dx$." Explain precisely what is wrong with this approach — there are two separate problems with it — and give the correct evaluation.

SOLUTION

Step 1. First problem with the student's approach: $dv = e^{x^2} dx$ has no elementary antiderivative at all, so " $v = e^{x^2}$ " is simply false — v must be an antiderivative of dv , and $\frac{d}{dx}[e^{x^2}] = 2xe^{x^2} \neq e^{x^2}$.

Step 2. Second, deeper problem: integration by parts is the wrong tool here regardless — the integrand xe^{x^2} is already in the exact form $f(g(x))g'(x)$ (with $g(x) = x^2$, $g'(x) = 2x$ up to a constant), which is a direct substitution, not a product needing parts.

Step 3. Correct evaluation: let $u = x^2$, $du = 2x dx$, i.e. $x dx = \frac{1}{2}du$: $\int xe^{x^2} dx = \frac{1}{2} \int e^u du = \frac{1}{2}e^u + C = \frac{1}{2}e^{x^2} + C$.

Common misconception: Reaching for integration by parts on *any* integrand that superficially looks like a product of two functions, without first checking whether it's actually a disguised chain-rule substitution — and without checking that the proposed v is a genuine antiderivative of dv .

19. (Level: Challenge — combined technique) Evaluate $\int e^{\sqrt{x}} dx$.

SOLUTION

Step 1. Substitute to remove the radical from the exponent: let $t = \sqrt{x}$, so $x = t^2$, $dx = 2t dt$: $\int e^{\sqrt{x}} dx = \int e^t \cdot 2t dt = 2 \int te^t dt$.

Step 2. The remaining integral needs a second technique, integration by parts: $u = t$, $dv = e^t dt \Rightarrow du = dt$, $v = e^t$: $\int te^t dt = te^t - \int e^t dt = te^t - e^t + C$.

Step 3. Combine and substitute back $t = \sqrt{x}$: $\int e^{\sqrt{x}} dx = 2(te^t - e^t) + C = 2e^t(t - 1) + C = 2e^{\sqrt{x}}(\sqrt{x} - 1) + C$.

Step 4. Verify by differentiating: $\frac{d}{dx} [2e^{\sqrt{x}}(\sqrt{x} - 1)] = 2 \left[e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} \cdot (\sqrt{x} - 1) + e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} \right] = \frac{e^{\sqrt{x}}}{\sqrt{x}} [(\sqrt{x} - 1) + 1] = \frac{e^{\sqrt{x}}}{\sqrt{x}} \cdot \sqrt{x} = e^{\sqrt{x}}$, matching the original integrand.

Common misconception: Trying integration by parts directly on $e^{\sqrt{x}}$ without first substituting away the radical in the exponent — parts alone cannot make progress while $\sqrt{x}$ sits inside the exponential.

20. (Level: Challenge — application) Water flows through a pipe at a rate $r(t) = 5te^{-t/2}$ cubic meters per minute at time t minutes, for $t \geq 0$. Find the total volume, in cubic meters, that flows through the pipe during the first 4 minutes, i.e. compute $\int_0^4 5te^{-t/2} dt$.

SOLUTION

Step 1. Choose $u = 5t$, $dv = e^{-t/2}dt$, so $du = 5 dt$, $v = -2e^{-t/2}$.

Step 2. Apply parts: $\int 5te^{-t/2} dt = -10te^{-t/2} - \int (-2e^{-t/2})(5) dt = -10te^{-t/2} + 10 \int e^{-t/2} dt$.

Step 3. Finish the remaining integral: $-10te^{-t/2} + 10(-2e^{-t/2}) + C = -10te^{-t/2} - 20e^{-t/2} + C = -10e^{-t/2}(t + 2) + C$.

Step 4. Evaluate from $t = 0$ to $t = 4$: at $t = 4$, $-10e^{-2}(6) = -60e^{-2}$; at $t = 0$, $-10e^0(2) = -20$. So $\int_0^4 5te^{-t/2} dt = -60e^{-2} - (-20) = 20 - 60e^{-2} \approx 20 - 8.12 \approx 11.88$ cubic meters.

Common misconception: Forgetting to evaluate the antiderivative at the *lower* limit $t = 0$ as well — since it isn't zero here ($-10e^0(0 + 2) = -20$, not 0), skipping it silently drops 20 cubic meters from the answer.